Beyond Screening Off: Transition Closure as the Dynamical Criterion for Rainforest Admission

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Abstract

Scale-relative structural realism allows that real patterns may occupy more than one level of ontology, but the admittance criteria for those levels remain underdetermined. Franklin and Robertson (2024) offer the most developed recent proposal: emergence, understood as screening off plus novelty, earns an entity its place in the rainforest. This paper argues that screening off is necessary but not sufficient. A macro-variable can enter into robust and novel macrodependencies that render micro-details conditionally irrelevant, yet still fail to carry autonomous macro-dynamics of its own. What is missing is a transition criterion. A candidate macro-object qualifies when, for a fixed regime, horizon, and admissible intervention class, macrostate information is sufficient for macro-transitions, so within-class micro-differences do not change macro-level what-follows. In exact Markov settings, strong lumpability provides a benchmark realization. In non-ideal settings, closure is graded through leakiness and convergent diagnostics under fixed constraints. The interventionist machinery is compatible with structural realism because interventions are used as diagnostic probes for relational stability, not as metaphysical primitives. The result completes rather than replaces existing admittance proposals: Dennett's compression identifies candidate patterns, screening off and novelty provide a serious filter, and closure determines which surviving candidates carry genuinely autonomous transition structure.

Keywords: real patterns, rainforest realism, closure, strong lumpability, coarse-graining, structural realism, interventionism

1 Introduction: Closure and the Admittance Problem

Scale-relative structural realism says that the world is not exhausted by a single privileged descriptive level. The special sciences succeed by tracking higher-level patterns that are stable enough to support prediction, explanation, and intervention. Ladyman and Ross's rainforest realism gave this thought its most influential metaphysical form: reality is populated by real patterns at many scales, not only by microphysical furniture (Ladyman and Ross 2007). But one question remains insufficiently settled. Which patterns should be admitted into that rainforest, and which should be treated more cautiously as useful summaries, temporary proxies, or dynamically idle aggregates?

Franklin and Robertson offer the most developed recent answer (Franklin and Robertson 2024). Their proposal is that emergence earns rainforest admission when two conditions are met: the macro-variable screens off micro-details relative to the explanandum, and the macro-variable contributes something genuinely novel at that level. That is a substantial advance. It captures why obvious gerrymanders and trout-turkey style composites should be excluded, while preserving the thought that higher-level ontology is earned by stable macro-level dependence rather than by intuitive naturalness alone. They explicitly cast the issue as one of the "admittance criteria for the rainforest" (Franklin and Robertson 2024, p. 3), and they do so in a way meant to remain compatible with theoretical reducibility rather than to revive a strong irreducibility test.

This paper argues that screening off is necessary but not sufficient. A macro-variable can satisfy observational screening-off conditions and enter into robust, law-like macrodependencies while still borrowing its apparent autonomy from upstream structure rather than instantiating autonomous macro-dynamics of its own. What is missing is a transition criterion. A candidate macro-object qualifies when, for a fixed regime, horizon, and admissible intervention class, macrostate information is sufficient for macro-transitions, so within-class micro-differences do not change macro-level what-follows. Screening off shows that micro-details are conditionally irrelevant for a target dependence. Closure asks the further question whether the candidate carries its own transition structure under admissible perturbation. That further test also sharpens a pressure already implicit in rainforest realism itself: if a macro-candidate is to be more than a physically allowed summary, it must be non-redundant in a dynamical sense, not merely useful or compressive.

The dialectical structure is cumulative rather than competitive. Dennett's real patterns identify candidate structures through compression and projectibility (Dennett 1991). Franklin and Robertson add a serious admittance filter by requiring screening off plus novelty (Franklin and Robertson 2024). The present paper argues that one further step is needed. Closure distinguishes the surviving candidates that are genuinely autonomous from those that remain parasitic on hidden micro-identity or a shared upstream cause. The aim is therefore not to replace rainforest realism, but to complete it by supplying the missing dynamical downgrade condition.

A contrast makes the gap vivid. Lewis observed that we have no principled reason to deny existence to the mereological sum of "the right half of my left shoe plus the Moon plus the sum of all Her Majesty's ear-rings," however sensible it is to ignore such things in ordinary thought (Lewis 1986, p. 213). That is correct as far as it goes.

But consider the difference between that composite and a hurricane. Knowing a hurricane's pressure organization and rotational structure tells you what it will do next without tracking every molecule. The shoe-moon composite carries no comparable transition structure; predicting its future requires independently tracking each component. Franklin and Robertson's criterion helps explain why the hurricane belongs in the rainforest while the shoe-moon composite does not. The argument here is that one can still construct macro-variables that survive screening-off tests while failing transition autonomy. The common-cause case in §3.3 is designed to show exactly that residual gap.

Throughout this paper, *gerrymandered* means dynamically incoherent: a partition that fails transition autonomy by requiring persistent within-class micro-bookkeeping. It does not mean merely intuitively disjunctive or aesthetically odd. By the same token, *parasitic* means that a macro-variable's apparent autonomy is borrowed from upstream structure rather than realized in the candidate partition's own transition organization.

This thesis is meant as a completion rather than a replacement. Existing pattern realism captures compression and projectibility insights. The present contribution adds a discriminating condition that excludes parasitic survivors by transition structure rather than by intuitive naturalness. The formal benchmark for this condition is strong lumpability in exact Markov settings. The non-ideal extension is graded closure via leakiness and convergent diagnostics under fixed constraints. The larger metaphysical payoff is that rainforest realism can then be read as a structured space of computationally closed coarse-grainings rather than as a binary inventory of what is, or is not, real.

The argument is philosophical, not a methods paper in disguise. The paper does not claim to deliver a universal estimation recipe. It provides a criterion and a disciplined protocol for applying it. Estimator selection, finite-sample behavior, and domain-specific implementation remain downstream methodological tasks.

1.1 Novelty and Positioning

The novelty claim has three parts.

1. Beyond screening off alone: Franklin and Robertson's criterion is retained as a necessary filter, but closure adds an intervention- indexed transition test that distinguishes autonomous macro-variables from parasitic ones. Their own account is valuable precisely because it frames rainforest admission without treating reducibility by itself as disqualifying (Franklin and Robertson 2024).
2. Beyond formal closure results alone: recent work distinguishes informational, causal, and computational closure as formal diagnostics (Rosas et al. 2024), but does not say when those diagnostics warrant ontological commitment or when commitment should be withdrawn. The present paper supplies that missing philosophical layer: an objecthood criterion with explicit intervention indexing, graded verdicts, and downgrade conditions.
3. Beyond effective realism without downgrade discipline: recent work in ontic structural realism allows regime-dependent, scale-relative ontology (Ladyman and

Lorenzetti 2024), but leaves open what operationally distinguishes robust admission from cases that should be downgraded or withheld. Closure supplies that discipline.

This is a conditional metaphysical proposal. Under structural realist and interventionist commitments, closure under admissible conditions is sufficient for macro-objecthood in regime. Readers who withhold those commitments can still accept a narrower conclusion: closure is at least a necessary anti-gerrymandering constraint on serious macro-ontology claims.

This positioning is deliberately dialogical. With Dennett, the paper keeps compression realism while refusing to let compression settle admittance by itself (Dennett 1991). With Franklin and Robertson, it agrees that screening off and novelty are indispensable, but argues that they do not yet separate autonomous from parasitic macro-variables (Franklin and Robertson 2024). With Ladyman and Ross, it keeps the rainforest ambition while insisting on an explicit transition criterion for objecthood verdicts (Ladyman and Ross 2007). With Ladyman and Lorenzetti, it treats interventions as diagnostics of structural autonomy rather than as metaphysical primitives (Ladyman and Lorenzetti 2024). Their language of the scale relativity of ontology helps make clear why this does not collapse realism into convenience: ontology can be regime-dependent while remaining mind-independent and answerable to objective modal structure (Ladyman and Lorenzetti 2024). With Rosas et al., the relationship is tool-user rather than commentary: their formal closure diagnostics supply measurement machinery, while the present paper supplies the philosophical criterion that determines what those measurements mean for ontology (Rosas et al. 2024). With causal-emergence work, it treats macro-level gains as corroborating diagnostics rather than as a standalone ontology test (Hoel et al. 2013; Dewhurst 2021). With Kim-style exclusion pressure, it defends non-redundancy at explanatory and control levels without denying microphysical implementation (Kim 1998). One consequence of intervention indexing is that the present criterion can diverge from purely observational diagnostics: a macro-variable can achieve predictive sufficiency by tracking a common cause rather than by instantiating autonomous transition structure, passing screening-off style tests while failing under admissible intervention (§3.3).

Three nearby traditions further sharpen the paper's posture. Simon's account of hierarchy and near-decomposability explains why complex systems can display relatively autonomous short-run subsystem dynamics alongside looser longer-run interdependence (Simon 1962; Ando and Simon 1961). That is a structural precursor to the regime- and horizon-indexed treatment of closure defended here. But Simon asks when aggregation is dynamically serviceable, not when such serviceability warrants ontological commitment.

Wimsatt's analysis of aggregativity marks a second antecedent. He argues that genuinely aggregative system properties are exceptional because most higher-level properties depend on organization and therefore fail invariance conditions such as unrestricted rearrangement, reaggregation, and intersubstitution of parts (Wimsatt 2000). That diagnosis fits the paper's anti-gerrymandering aim. But non-aggregativity alone does not show that a candidate macro-description carries self-contained transition structure. Closure adds that further test.

The mechanisms tradition provides a third point of contact. Machamer, Darden, and Craver characterize mechanisms in terms of organized entities and activities productive of regular change, while Craver's later account emphasizes that explanatory levels are local to mechanisms rather than pieces of one global hierarchy (Machamer et al. 2000; Craver 2007). The present paper is continuous with that tradition, but adds a criterion for when a candidate variable or coarse-graining has earned a place in a mechanism-level ontology at all.

Several recent contributions make the present dialectical setting more precise. Millhouse tightens Dennett's realism without yet requiring transition autonomy (Millhouse 2022). He is the clearest nearby case of a stronger real-pattern view that still stops short of the transition test defended here, because predictive success and high-level mapping can remain too permissive without a stricter dynamical condition (Millhouse 2022). Wallace presses the discussion toward patterns in dynamics and distributions rather than mere collections of parts (Wallace 2024). Jiang's work in biology and mechanisms shows that closely related ontological questions are now being asked across multiple domains (Jiang 2024, 2025). In mechanisms, Jiang explicitly uses structural realism to press against object-based metaphysics; in biology, the same broad terrain is framed more permissively in terms of explanatory and pragmatic utility (Jiang 2024, 2025). The recent volume on real patterns, together with Ladyman's contribution to it, confirms that the dynamic and structural interpretation of pattern realism is now an explicit topic rather than a background implication (Millhouse et al. 2026; Ladyman 2026). The present paper's narrower claim is that dynamic or phase-space formulations still leave open an admittance and downgrade question. Closure is proposed as that further discipline, not as a rival to the move toward dynamic structure.

A related line of work traces the cognitive origins of real-pattern commitments, linking predictive coding and free-energy minimization to how bounded agents build and revise their representational schemes (Gładziejewski 2025). That work helps explain why agents converge on some macro-descriptions and abandon others, which strengthens the anti-arbitrariness side of pattern realism. But representational success can remain task-relative unless a further test determines when a candidate grouping tracks dynamical structure in the world rather than merely serving local inferential needs. The closure criterion supplies that additional test.

A stronger reading of Dennett is worth confronting directly. One can interpret real-pattern realism as already requiring projectibility and cross-context robustness, not merely compression in a single dataset. On that steelman, Dennett's criterion already excludes fragile, narrowly tuned codings. Even so, projectibility constrains which predictions are worth tracking; it does not test whether the macro-description carries autonomous transition structure. A partition can be projectible across contexts and still require persistent within-class micro-bookkeeping to sustain its forecasts, because the macro-label groups together microstates whose onward profiles diverge once perturbation or horizon extension is applied. Closure catches exactly that residual failure mode. It asks not only whether a pattern projects, but whether onward macro-transitions are self-contained at the declared grain. That is the gap this paper claims remains even after the best current screening-off proposals.

In short, projectibility and screening off can still tolerate successful forecasting that depends on recurring micro-level repair. Closure does not.

1.2 Intervention, Structure, and the Causation Worry

One immediate tension should be stated early. Ladyman and Ross are skeptical that causation belongs in fundamental metaphysics, and recent effective ontic structural realism continues that broadly structuralist posture (Ladyman and Ross 2007; Ladyman and Lorenzetti 2024). By contrast, the criterion defended here is explicitly intervention-indexed. That can look like a retreat from structural realism to a Woodwardian metaphysics of causation. It is not.

Intervention does not function here as an ontological primitive. It functions as a diagnostic probe for structural autonomy. The claim is not that the world is fundamentally made of interventions, but that admissible intervention is our most disciplined test of whether a candidate macro-partition preserves stable relational organization under perturbation. What the criterion tracks is still structure. The intervention test asks whether that structure is genuinely autonomous at the candidate grain or whether the apparent autonomy was borrowed from elsewhere.

This is why the proposal is compatible with effective ontic structural realism. Ladyman and Lorenzetti explicitly allow ontology to be regime-dependent and scale-relative while remaining realist about the structures discovered there (Ladyman and Lorenzetti 2024). The present paper agrees, but adds a sharper admittance condition: a scale-relative structure earns robust ontological standing when its transition profile is stable under the admissible perturbations that define the regime. In that sense, intervention is used here to supply operational discipline for effective realism rather than to compete with it. Their claim is not that ontology collapses into convenience, but that ontology is always effective ontology at some scale and under some modal profile (Ladyman and Lorenzetti 2024); the present paper adds a stricter account of when such scale-relative commitment should be robust, qualified, or withheld.

1.3 Scope and Non-Claims

A preliminary distinction is needed. Closure can be stipulated or induced. In stipulated closure, autonomy is fixed by constitutive rules, as in chess or formal arithmetic: once the rules are set, transitions and consequences follow by construction. In induced closure, autonomy is discovered through dynamical screening-off in the physical world. This paper is concerned exclusively with induced closure in spatiotemporal systems. It does not offer a complete metaphysics of levels, and it does not settle all questions about abstract objects or full mereology. Where those debates arise, they are handled only to protect the central claim from predictable misreadings. Still, the criterion is stated over state-transition structure: spatiotemporal dynamics are the paradigm case, not the only intelligible format in which closure questions can be posed.

Methodologically, the paper does not depend on the strongest anti-analytic rhetoric sometimes associated with naturalized metaphysics. The claim is narrower: closure is a usable philosophical constraint that can be motivated by dynamical and coarse-graining practice, whether or not one accepts every meta-metaphysical commitment in the broader *Every Thing Must Go* (ETMG) program (Ladyman and Ross 2007).

The roadmap is direct. Section 2 states the criterion and admissibility discipline, and shows how closure sharpens the non-redundancy demand already implicit in the Primacy of Physics Constraint. Section 3 gives the exact benchmark and draws out the lattice payoff of computationally closed coarse-grainings. Section 4 extends the criterion to non-ideal settings and explains the graded verdict structure. Section 5 addresses the strongest objections. Section 6 summarizes gains and limits.

1.4 Three Nearby Positions

It helps to separate three nearby stances that are often run together.

1. Compression-prediction realism: successful compression and forecasting are treated as sufficient for realist commitment.
2. Pure pragmatism: model choice is guided by utility alone, with no ontological consequence.
3. Closure realism: compression and prediction matter, but on the present account objecthood requires transition autonomy under fixed regime, horizon, and admissible intervention class.

The third stance preserves the explanatory strengths of the first while avoiding its permissiveness. It also avoids reducing ontology to convenience, which is the pressure on the second.

1.5 Contribution Map

The paper makes four linked contributions.

1. It gives a criterion-level tightening of real-pattern realism by adding closure as a discriminating objecthood condition.
2. It provides a bridge from exact benchmark cases to non-ideal cases without changing criterion content.
3. It offers a disciplined verdict structure with explicit downgrade conditions, so realism claims remain answerable to failure.
4. It shows that closure supplies the dynamical condition the Putnam-Fodor autonomy tradition always needed but left implicit: multiple realizability groups realizers extensionally, while closure demands that those realizers produce the same macro-transitions (§5.4).

The central payoff is selective commitment. The account can support strong commitments where transition autonomy is robust, while avoiding forced commitments where evidence is fragile or regime-sensitive.

2 Closure and Admissibility

2.1 Criterion in Plain Language

Closure can be stated without heavy formalism. A candidate macro-object passes when knowing its current macrostate is enough to determine its macro-future over a specified horizon and intervention class. If two microstates inside one macrostate produce different macro-transitions, closure fails at that grain.

Static patterns are not exceptions to this rule. They represent dynamical regimes where micro-fluctuations are screened off rather than absent.

Persistence is therefore a particular type of macro-transition that falls squarely within the closure criterion.

The key point is dynamic. The criterion concerns transition sufficiency, not mere descriptive fit. A representation can summarize trajectories elegantly and still fail objecthood if it requires persistent within-class micro-bookkeeping to preserve forecast quality.

Since the formal machinery evaluates partitions of state spaces rather than physical things directly, an explicit bridge is needed. On the structural realist commitments defended below (§2.11), a macro-object just is stable relational and causal organization at the relevant grain; the partition identifies the object by identifying that organization.

2.2 Formal Statement

Let micro-process be X_t and candidate macro-process be $Z_t = g(X_t)$. For horizon L, closure asks whether Z_t is sufficient for forecasting Z_{t+1}^L under a fixed regime and admissible intervention class.

The horizon object here is the L-step path distribution, not only one-step prediction. One-step tests can be used as diagnostics, but objecthood claims are indexed to the declared horizon target.

Informationally, the predictive side asks whether adding X_t beyond Z_t yields material gain for macro-future prediction. Interventionally, the causal side asks whether macro-transition laws remain stable across admissible perturbations without needing within-class micro-identification.

These are not competing tests. They are two readings of the same target: transition autonomy at the macro level.

2.3 Why This is More Than Compression

Compression is necessary but not sufficient for objecthood. A compressed code can hide transition-relevant heterogeneity and still score well on narrow tasks. Closure blocks that loophole by asking whether hidden heterogeneity matters for onward macro-behavior.

Closure therefore functions as an anti-gerrymandering condition. It does not reward convenient coding alone. It rewards coarse-grainings that carry stable transition structure under declared constraints.

Put differently, this paper accepts Dennett's claim that compression success is evidence of objective patterning, but denies that compression success is the full ontological test (Dennett 1991). It also accepts Ladyman and Ross's concern that real patterns should be projectible and structurally disciplined, while adding an explicit transition criterion for adjudicating borderline cases (Ladyman and Ross 2007). The aim is not to replace those insights. The aim is to make their realist force less permissive.

This also answers a familiar Haugeland-style pressure point. The paper does not infer objecthood directly from "there is a useful pattern in the data." It adds an explicit transition-autonomy condition that determines when pattern talk has ontological force rather than merely descriptive convenience (Haugeland 1998).

2.4 Closure and the Primacy of Physics Constraint

Ladyman and Ross's Primacy of Physics Constraint is often read negatively, as a warning against ontologies that float free of physics. But it also contains a second pressure that matters just as much here. Macro-candidates must not only be physically compatible. They must also avoid redundancy with lower-level description in the realism-relevant sense (Ladyman and Ross 2007). The missing question is what that non-redundancy amounts to once one gives up the idea that only the micro-level is fully real.

The present proposal is that closure gives that vague requirement operational content. A macro-candidate is non-redundant when micro-differences within one macroclass do not alter macro-transitions under the declared regime, horizon, and admissible intervention class. That is a stronger claim than descriptive usefulness, and a different claim from mere compression. A macro-description can be shorter, more tractable, and even explanatorily attractive while still remaining redundant in the sense that its apparent autonomy depends on continued micro-level repair.

This is why the common-cause pressure case matters. A proxy can be fully compatible with physics and can look projectible under passive observation because upstream structure keeps its trajectories aligned. Yet the same proxy can remain transition-parasitic: once admissible intervention perturbs the candidate directly, within-class micro-differences matter again and the apparent macro-autonomy disappears. Closure therefore sharpens the Primacy of Physics Constraint's non-redundancy demand by making it a question about transition structure rather than about descriptive convenience.

Read this way, closure is not an external supplement imposed on rainforest realism. It is a way of saying more exactly what it takes for a higher-level pattern to be physically respectable without collapsing back into lower-level bookkeeping. The realism-relevant discriminator is causal autonomy under admissible perturbation, not microphysical smallness by itself.

2.5 Admissibility and Hierarchical Evaluation

Closure is always relative to an intervention class. The standard objection is that this invites circularity: who picks the class? The foundational answer is that the system's own transition dynamics do. A regime is a region of dynamical possibility carved out by the transitions themselves, and the admissible intervention class is the set of perturbations that regime tolerates without dissolving. The analyst's role is to read off that class honestly, not to invent it. Procedural discipline (below) ensures the reading is auditable, but the source of the constraint is the dynamics, not the analyst.

Objecthood, on this view, is stability of macro-transition structure across the space of perturbations the regime actually sustains, not merely fit to observed trajectories. The constraints below make this space explicit and physically anchored.

Admissibility is fixed upstream of objecthood verdicts by three constraints:

1. Epistemic admissibility: interventions are measurable and implementable by bounded agents.
2. Dynamical admissibility: interventions preserve the target regime class.
3. Explanatory admissibility: interventions target variables with cross-context generalizability and non-trivial counterfactual reach, rather than one-off manipulations tuned to a single episode.

These constraints are physically anchored in available control channels and implementation structure, not in analyst preferences about favored ontologies. The explanatory constraint is set before closure verdicts and does not assume in advance which candidate partition will prove stable. This makes the interventionist notion of a "possible experiment" concrete and bounded, rather than leaving it at conceptual possibility where nearly any hypothetical manipulation might count (Woodward 2015). It also aligns with Woodward (2021)'s argument that correct variable choice requires satisfying "proportionality": minimizing the omission of dependency relations. Domain expertise still shapes regime identification and control-channel selection, but that shaping is declared, auditable, and separated from partition scoring.

Operationally, explanatory admissibility can be audited without looking at which partition wins. A candidate intervention class is stronger when it is implemented through control channels that already exist in the domain and are replicable across sites, operators, or runs. It is weaker when success depends on narrow, one-off tuning that fails under small, feasible perturbations of the same intervention family.

A related distinction matters here. Pearl defends the $do(x)$ operator as an ideal "surgery" that can be defined mathematically regardless of physical feasibility (Pearl 2009). For causal inference, that idealization is productive: it lets one reason about effects even when direct manipulation is unavailable. But macro-objecthood, on the view defended here, requires tighter constraints than logical derivation. A partition that achieves closure only under physically unrealizable interventions, ones that no bounded agent could implement without destroying the target regime, does not support the kind of stable, intervention-guiding structure that warrants realist commitment. The admissibility constraints above ensure that closure is assessed relative to control channels that are actually available in the regime, not relative to arbitrary mathematical surgeries. This is not a rejection of Pearl's framework. It is a restriction on which of its outputs carry ontological weight.

This point connects back to the foundational claim above. While we often speak of “interventions” as laboratory actions performed by human scientists, the admissible class is ultimately defined by the objective physical ecology of the system. Nature intervenes on itself. A system’s boundaries are constantly probed by naturally occurring environmental perturbations: thermal noise, chemical diffusion gradients, and mechanical impacts. A cell membrane objectively screens out certain molecular fluctuations whether a human experimenter is observing it or not. The transition dynamics of the cell carve out a regime, and that regime determines which perturbations are admissible: those the system processes or resists while maintaining its macro-transition structure. The intervention class is therefore not a human invention; it is a formalization of the space of possibility the dynamics already define.

Evaluation order matters.

1. Fix regime, horizon, admissible intervention class, diagnostics, and model class.
2. Compare candidate partitions under those fixed constraints.
3. If constraints are revised after inspecting results, restart and disclose the revision.

This order blocks back-fitting and makes disagreement tractable. If verdicts remain highly sensitive across nearby defensible admissibility specifications, commitment should be downgraded. Stated as a discipline, admissibility has two stages: first, intervention classes are fixed from independent physical constraints and established control practice in the target domain; second, closure is evaluated only relative to that fixed class.

2.6 Regime Individuation and Circularity

A structural worry deserves explicit treatment here. If regime individuation already commits to which macro-variables matter, the protocol risks a version of circularity: closure testing presupposes regime identification, but regime identification might presuppose partition outcomes. The response is that defensible regime specifications draw on quantities characterizable without presupposing which candidate partition will succeed. Temporal parameters, energy scales, flow rates, and enforcement intensities are typically identifiable through physical indicators that competing candidate partitions can agree on before either is evaluated. In the traffic illustration, “weekday commuter traffic in one metropolitan corridor” is characterized by clock schedules, total vehicle counts, and road network topology, all quantities that investigators favoring either the lane-flow or vehicle-ID-cluster partition can agree on before either closure test is run. Regime specifications of this kind are anchored in lower-level or cross-cutting physical indicators rather than in the outcome of the very test they enable. The key principle is that regime specifications should be grain-neutral across competing candidate partitions: a regime can itself be a macro-level description while still not presupposing which finer-grained partition will close within it.

The independence is imperfect in some domains, particularly biology and social science, where macro-variable choice and regime characterization can be genuinely entangled. The present account treats that entanglement as a source of evidential weakness rather than a concealed defect. When alternative, independently motivated regime specifications produce divergent closure verdicts, the verdict ceiling should be downgraded to at most qualified. More broadly, regime individuation and partition evaluation are iterative rather than viciously circular. Initial regime characterizations are provisional, and closure testing can refine them. The process stabilizes into reflective equilibrium in the sense Goodman articulated: rules of inference and particular inferential judgments are justified by mutual coherence rather than by either serving as the unilateral foundation for the other (Goodman 1983). If iteration does not stabilize across nearby defensible starting specifications, the result should be treated as indeterminate for that regime rather than forced into a robust verdict. Here stabilization means that comparative ordering across independently specified candidate partitions stops shifting under small, defensible protocol variations; it does not presuppose one privileged grain in advance. The discipline comes from requiring that all revisions be disclosed and that each revision restart the evaluation rather than silently inheriting prior results.

2.7 Pattern Reality Versus Macro-Objecthood

One distinction is essential for avoiding confusion. A pattern can be real in a weaker sense without satisfying this paper's objecthood criterion. A representation can capture regularities that are descriptively useful while still failing closure under admissible interventions.

The criterion in this paper is stricter. It targets macro-objecthood, not any and every projectible summary. The paper can therefore preserve a generous attitude toward pattern detection while denying ontological standing to high-leak candidates.

That distinction also clarifies the paper's relation to Franklin and Robertson. Their screening-off condition is well suited to identifying when a macro-variable has become explanatorily and nomically serious (Franklin and Robertson 2024). The present claim is narrower and stricter: some variables that are serious enough to survive that filter still fail transition autonomy and so do not yet qualify for robust macro-objecthood.

The same distinction also clarifies what this view contributes to composition debates. In van Inwagen's terms, the standing question is when some things compose a further thing (van Inwagen 1990). The present proposal does not offer a full mereology. It gives a domain-indexed discipline condition: composition claims about macro-objects are warranted when transition autonomy is robust under declared constraints, and should be withheld when candidate sums remain transition-fragile.

That distinction clarifies dialectical burden. The paper does not need to show that non-closed representations are worthless. It only needs to show that they do not meet the objecthood standard defended here.

2.8 Admissibility Disputes

Admissibility disputes are expected, especially across domains. The adjudication rule is to compare candidate intervention classes by what they physically permit and whether they preserve the same target regime. Boundary-pressure perturbations can be admissible for storm dynamics, while molecule-by-molecule remote rewriting is not. In institutional settings, changing enforcement intensity can be admissible, while instant arbitrary rewriting of all agent commitments is not. The ontological consequence is direct. A storm satisfies closure under boundary perturbations because its pressure organization and rotational structure determine macro-transitions without tracking individual molecules. Under molecule-by-molecule rewriting, the same candidate fails, because admissibility now permits interventions that reach inside the macroclass and exploit within-class differences. The candidate has not changed; the admissible control channel has, and the verdict changes with it. Admissibility must therefore be declared and justified before scoring, not adjusted afterward to protect a preferred outcome.

For transparency, three template classes are often useful as a starting grid: boundary nudges, coarse actuator controls, and policy levers. These templates are not universal, but they make admissibility comparison public and repeatable.

One guardrail is non-negotiable. Interventions that directly target preservation of partition labels, rather than system variables, are inadmissible because they trivialize closure by design. The test concerns whether transition structure is autonomous under physically meaningful controls, not whether labels can be protected by stipulation.

When two intervention classes are both admissible and target the same regime, the criterion permits plural testing rather than forced monism. Verdicts should report which admissibility class was used and how sensitive results are across nearby defensible classes. A candidate partition that appears closed only under one narrowly tuned class should be downgraded when sensitivity appears across nearby defensible alternatives.

2.9 Canonical Criterion Statement

For ease of reference, the core criterion can be stated compactly:

A candidate coarse-graining $Z = g(X)$ qualifies for macro-objecthood in regime when, under fixed horizon L and admissible intervention class $\mathcal{I}$, macro-transition structure is autonomous up to declared tolerance, so within-class micro-differences do not materially improve macro-future prediction or intervention-guided control.

Candidate partitions must be specified independently of the particular sample trajectory used to score closure, for instance by a pre-declared construction rule or learning objective fixed before evaluation. Otherwise closure collapses into bespoke encoding, where any dataset can be made to look autonomous by post hoc recoding.

In exact Markov settings, strong lumpability is a sufficient benchmark realization of this criterion. In non-ideal settings, leakiness-centered diagnostics estimate distance from that ideal under fixed constraints.

2.10 Predictive and Interventional Closure

Predictive and interventional closure should be distinguished but not separated. Predictive closure concerns whether macrostate information screens off transition-relevant micro-detail for macro-future forecasting. Interventional closure concerns whether macro-transition structure remains stable under admissible perturbation.

The evidential relation is asymmetric. Strong interventional closure typically implies predictive adequacy for the same target and horizon, while predictive adequacy alone can persist in cases where interventional stability fails under distribution shift. For that reason, the criterion treats predictive evidence as important but not final.

This asymmetry also clarifies burden of proof. Claims to robust macro-objecthood need either direct interventional support or compelling indirect evidence that tracks intervention-relevant invariance. Otherwise, the responsible verdict is qualified or indeterminate.

2.11 Why Closure Carries Ontological Weight

A reviewer can still ask why transition autonomy should count as objecthood rather than as a mere success condition for modeling. The answer depends on the paper's explicit commitments. Under structural realism, what ontology should track is stable relational and causal organization rather than intrinsic micro-identity as such. Under interventionism, what counts as causally relevant structure is structure that supports stable manipulation and control.

This fits the regime-dependent, scale-relative orientation defended by effective ontic structural realism. The present proposal does not ask for a return to one privileged level or to thick intrinsic objects. It asks for a sharper criterion of when a scale-relative structure has earned commitment, and when it should be downgraded, within a broadly structural realist framework (Ladyman and Lorenzetti 2024).

The meta-philosophical stance here follows what Woodward calls the methodological path to ontology: causal-interventionist tools are used to determine what earns realist commitment, rather than seeking independent metaphysical "truth-makers" or "grounds" that stand apart from the practices of manipulation and prediction (Woodward 2015). This is not a retreat to instrumentalism. It is the claim that scientific ontology, disciplined by closure, is the only ontology required for macro-objecthood verdicts. The demand for a further metaphysical foundation beyond stable dynamical autonomy under admissible intervention mistakes a philosophical habit for a substantive requirement.

A persistent metaphysical worry is that prediction and interventional control are merely epistemic lenses, leaving the underlying ontology untouched. This confuses the test with what is being tested. Prediction and intervention are diagnostic tools that reveal pre-existing statistical structure. The conditional independencies that allow a macro-state to screen off micro-details exist whether or not anyone computes them; they are the terrain, not the lens. Locality, finite signal propagation, and thermodynamic limits create these independencies as physical facts. Our predictive models succeed when they align with this objective causal topology, and fail when they do not. Closure is therefore an ontological criterion because it identifies where the world's phase space objectively decouples, not just where our minds prefer to chunk it.

Given those commitments, closure does not function as a convenient heuristic layered on top of ontology. It is the criterion that identifies when a candidate macro-description tracks stable structure at the level where explanation and intervention are being assessed. This is why the view can remain compatible with microphysical completeness while still defending non-redundant macro commitment.

That point has a further metaphysical consequence. If one asks what the rainforest amounts to once admission is disciplined this way, the best answer is not a loose plurality of levels but the ordered space of computationally closed coarse-grainings. Rosas et al. provide the formal architecture for talking about such relations among partitions (Rosas et al. 2024). The present paper's claim is philosophical: lattice membership is what it is for a candidate level to become ontologically serious. Admission, qualification, and downgrade are therefore not floating labels imposed from outside the formalism. They locate a candidate within a structured space of more and less stable closure relations.

2.12 Structural Contrasts

The connection to Pearl's notion of modularity is worth noting. Pearl argues that causal models work because nature consists of autonomous mechanisms: changing one structural equation need not disrupt the others (Pearl 2009). Closure is related to this property but not identical to it. Pearl treats modularity as a structural assumption about a given causal model; the present criterion asks a prior question, namely whether a candidate coarse-graining has earned the right to appear as a variable in such a model at all. A partition that requires persistent micro-bookkeeping to preserve its transition structure has not isolated an autonomous mechanism at the macro grain in the relevant sense. So closure can be understood as an ontological precondition for the kind of modularity Pearl's framework presupposes, rather than as a restatement of it.

The proposal is accordingly not vulnerable to standard "bare structure" worries. The criterion is modal and dynamical, not merely extensional. It is about counterfactual transition organization under interventions, not about abstract isomorphism alone.

The same point clarifies the relation to mechanistic views of levels. Mechanistic accounts are right that organized entities and activities can support stable explanation and intervention (Machamer et al. 2000; Craver 2007). But by themselves they do not settle whether a proposed macro-variable has been carved at the right place. Closure addresses that prior sorting question. A candidate may be heuristically useful or mechanistically suggestive and still fail objecthood if admissible perturbation reveals continued dependence on transition-relevant microstructure.

This also addresses Newman-style triviality pressure in structural realism, where purely extensional structure can be cheap under redescription (Ainsworth 2009, pp. 141–145). Closure is not extensional fit alone. It requires transition and intervention stability under declared constraints, which arbitrary recodings typically fail. When apparent success depends on privileged representational accidents, that encoding dependence is itself evidence of leakiness.

Why not stop at "good variable" language? Because a good variable can be merely convenient, and the worry is that closure only tracks inferential convenience rather than what is objectively there. The present account rejects that deflation because closure claims are indexed to intervention classes, not only to passive prediction. A representation can score well on retrospective fit by exploiting accidental correlations and still fail when admissible interventions shift transition pathways. A closure-supporting partition survives those shifts because the screened-off structure is not accidental. If a candidate supports reliable counterfactual control and regime-stable transition laws, it is tracking objective modal structure: stable counterfactual dependencies and lawlike transition regularities under admissible intervention. On the paper's commitments, treating that structure as merely representational would undercut the very realism the criterion is designed to preserve. The modal profile, not the in-sample fit, carries the ontological burden. Predictive sufficiency can reveal compression, but interventional stability identifies difference-makers: it tests whether the same macro-transition structure survives admissible manipulation rather than merely fitting observed trajectories.

The view therefore distinguishes epistemic humility from ontological deflation. Finite agents will often have partial, noisy, and regime-limited evidence. But that uncertainty does not collapse the difference between dynamically autonomous and dynamically incoherent partitions. That difference can be difficult to estimate, but it is still a difference in the world rather than in preference.

2.13 Levels of Claim and Representation

Some recurring misunderstandings come from sliding between different levels of claim. The paper uses the following distinction throughout.

1. World dynamics: the implemented process itself.
2. Pattern type: stable transition structure supported by that process.
3. Pattern token: a concrete instance under specific boundary conditions.
4. Representation: a model, equation, classifier, or narrative that tracks the pattern.

A representation can fail while the pattern type remains real. A token can fail while the type remains robust. A token can also succeed while the type is weak, for example in narrow calibration windows. Closure claims in this paper are primarily type-level claims under specified regimes, then secondarily claims about token reliability under perturbation.

The criterion therefore does not equate model performance with ontology. It asks whether the represented transition structure is genuinely autonomous, not whether one representation currently performs well.

2.14 Closure, Underdetermination, and Objecthood Discipline

One remaining concern is underdetermination. Even after the criterion is stated, a reviewer may argue that many distinct coarse-grainings can be tuned to look acceptable on available data. If so, closure might seem to collapse back into model choice rather than objecthood discipline.

The answer is to separate three questions that are often conflated.

1. Which candidate partitions are descriptively serviceable in a dataset?
2. Which candidates remain transition-autonomous under admissible perturbation?
3. Which of those candidates remain stable under modest horizon and admissibility variation?

Underdetermination is strongest at the first question and weaker at the second. It is often weakest at the third. Many candidates can fit observed trajectories. Far fewer preserve counterfactual structure when the regime is probed. Fewer still remain stable across nearby defensible setups. This staged filtering is exactly where closure earns its metaphysical role.

The account does not claim that every domain yields one uniquely privileged partition. It claims that objecthood commitments should be constrained by transition autonomy and robustness, rather than by fit alone. When underdetermination persists after those filters, the responsible result is qualified or plural commitment under transparent constraints, not forced monism and not unconstrained relativism.

This is also the right place to state the lattice payoff directly. Once admissibility, horizon, and intervention class are fixed, the question is not simply whether one favored partition closes. The deeper question is which coarse-grainings close, which close only conditionally, and which fail. That structured answer is what the rainforest becomes under a closure discipline: the lattice of computationally closed coarse-grainings, ordered by refinement and by comparative stability. Levels on this view are discovered features of system dynamics rather than metaphors of convenience. They are nested by coarse-graining relations rather than eliminatively ranked. And because the relevant criterion is causal autonomy, not implementation grain, the micro-level is not privileged simply by being smallest.

Before moving to the formal benchmark, the practical upshot is simple: when exact closure is unavailable, leakiness-centered diagnostics are used to estimate comparative distance from closure, and commitment strength tracks the stability of that evidence.

3 Exact Benchmark: Strong Lumpability

Strong lumpability is introduced as a benchmark, not as destiny. It gives a clean exact case in Markov settings where closure can be stated and checked without ambiguity.

Let partition cells under g be macroclasses. Strong lumpability holds when microstates within the same macroclass induce identical transition probabilities to all macroclasses (Kemeny and Snell 1960). When this condition holds, macro-transitions are autonomous by construction.

More precisely: for any two microstates x, x' in macroclass C_i, and any macroclass C_j, strong lumpability requires $\sum_{y \in C_j} P(y \mid x) = \sum_{y \in C_j} P(y \mid x')$.

Lemma. If the micro-process is first-order Markov and the partition induced by g is strongly lumpable, then the induced macro-process is itself Markov and transition-autonomous at the macro level.

Proof sketch. Define a macro-kernel by summing micro-transition probabilities from any representative $x \in C_i$ into each macroclass C_j. Strong lumpability guarantees this sum is representative-independent. The macro-kernel is therefore well-defined, and the induced process over macroclasses is Markov with transitions given by that kernel.

The philosophical significance is straightforward. A partition that satisfies this condition is not merely useful. It preserves transition structure at the macro grain. A partition that fails it mixes transition-heterogeneous microstates and therefore lacks macro autonomy.

Equivalent language from computational mechanics clarifies why this is not superficial bookkeeping. For any macro-process Z, two prediction machines can be defined. The ε-machine is the minimal model that predicts Z's future from Z's own past, using only macro-level information. The υ-machine is the minimal model that predicts Z's future from the full micro-past X, using everything available (Shalizi and Crutchfield 2001; Rosas et al. 2024). Closure holds when these two machines are equivalent: the ε-machine already captures everything the υ-machine knows about macro-futures, and extra micro-information is redundant for the macro target. This gives an exact ideal where closure is not approximate.

The important limitation is equally clear. Exact strong lumpability is uncommon in open, noisy, and path-dependent systems. That limitation motivates the graded extension. It does not undermine the criterion.

3.1 Minimal Formal Illustration

The anti-gerrymandering role can be shown with a minimal symbolic example. Let microstates be $\{x_1, x_2, x_3, x_4\}$ with transition rows:

$$\begin{aligned}
P(x_1, \cdot) &= (\alpha, \beta, 0, 0),\\
P(x_2, \cdot) &= (\alpha, \beta, 0, 0),\\
P(x_3, \cdot) &= (0, 0, \gamma, \delta),\\
P(x_4, \cdot) &= (0, 0, \gamma, \delta),
\end{aligned}$$

with $\alpha + \beta = 1$ and $\gamma + \delta = 1$.

Partition A groups $\{x_1, x_2\}$ and $\{x_3, x_4\}$. Partition B groups $\{x_1, x_3\}$ and $\{x_2, x_4\}$. In A, members of each macroclass have matching onward profiles to macroclasses, so macro-transitions are autonomous. In B, members of one macroclass differ in onward profiles whenever $\alpha \neq \gamma$, so the macro-label hides a transition-relevant difference.

Partition B can still be made predictive by adding repeated within-class bookkeeping. A partition is *high-maintenance* when reliable macro-prediction repeatedly forces refinement of within-class distinctions, so that yesterday's macrostate labels must be supplemented by new micro-bookkeeping to retain accuracy. That is exactly the case this paper excludes from robust macro-objecthood. On a fixed sample trajectory, this bookkeeping can make Partition B look nearly as predictive as Partition A. The difference appears under admissible perturbation: A preserves macro-transition structure without renewed microstate tracking, whereas B does not.

The philosophical lesson is simple. Both partitions are definable. Only one preserves transition autonomy. Closure therefore discriminates between legitimate macro-candidates and merely codable aggregates.

3.2 If an Unusual Partition Closes

A common reaction is that a strange disjunctive partition might satisfy the formal condition in a symmetric system. That is correct. On this account, if such a partition genuinely supports autonomous macro-transitions under fixed constraints, it counts as real at that grain.

This is not a concession to arbitrariness. The criterion tracks transition coherence, not intuitive naturalness. If one wants a separate naturalness filter, that is an additional criterion and should be declared as such.

That result should also be read alongside Andersen's argument that structure may be dense rather than sparse (Andersen 2025). If multiple partitions close at different grains, that is not by itself an embarrassment for the view. It is what one should expect if the world's structural organization is richer than any single privileged carving. What matters is whether the candidate partition earns admission by closure under stated constraints, not whether it is the only possible closed partition.

The exclusion claim should therefore be read carefully. The criterion excludes dynamically incoherent, high-leak aggregates. It does not exclude every unusual grouping.

3.3 What the Formal Machinery Alone Does Not Provide

The mathematical property and the philosophical criterion are different things. Four elements of the present proposal have no counterpart in the formal literature alone.

First, intervention indexing. Causal-state constructions classify histories by equivalence of future distributions under the observed process, which is primarily an observational-predictive equivalence relation (Shalizi and Crutchfield 2001). The present criterion adds an admissible intervention class as a parameter of the closure test. Objecthood claims are indexed to what the system does under perturbation, not only under passive observation. This is what separates the criterion from a purely statistical diagnostic. Rosas et al.'s framework distinguishes informational, causal, and computational closure as complementary formal diagnostics (Rosas et al. 2024). They explicitly describe the result as a "lattice of nested computational structures" (Rosas et al. 2024, p. 2). The present paper treats predictive sufficiency as evidential, but ties objecthood commitment to admissible-intervention stability and to the absence of hidden micro-identity bookkeeping.

Second, an explicit ontological interpretation. Strong lumpability tells you when a coarse-graining preserves transition structure. It does not tell you whether that preservation warrants realist commitment. The philosophical work is to argue that, under stated commitments, transition autonomy under admissible interventions is sufficient for macro-objecthood, and to give the conditions under which that claim should be downgraded or withdrawn.

Third, a graded commitment protocol. The formal literature offers exact conditions and, more recently, approximate-closure measures. It does not supply a protocol for translating those measures into warranted ontological verdicts with explicit robustness requirements and downgrade rules. The selective commitment structure (robust, qualified, indeterminate) is a philosophical addition. These are not merely epistemic confidence labels. They report where a candidate sits in the space of more and less stable closure relations: robust where closure survives modest admissibility and robustness checks, qualified where closure is real but regime-fragile or only partially secured, and indeterminate where the candidate's place in that structure is not yet stable enough to underwrite commitment.

Fourth, an anti-gerrymandering argument. The argument that closure is the right anti-gerrymandering condition, and that compression without closure is insufficient for objecthood, because compression can be won by codings that effectively preserve hidden micro-identities, is not a theorem. It is a philosophical claim defended by the structure of Sections 2 through 4.

A simple case shows the divergence concretely. An aggregate macro-variable can track a latent process so faithfully that it achieves observational predictive sufficiency: an ε-machine built on the proxy compresses past observations and predicts future ones as efficiently as the full-state representation. Classical computational mechanics would recognize it as identifying a real pattern. But under admissible intervention on the proxy itself, transition structure collapses, because the variable was tracking a common cause rather than instantiating an autonomous mechanism.

Consider a concrete instance. Suppose a composite variable aggregates two subsystems that are both driven by a shared upstream oscillator. Under passive observation, the composite's macro-state predicts its own future as well as the full micro-state does, because the shared driver synchronizes the subsystems and makes their joint trajectory compressible. An ε-machine built on the composite would find no residual micro-information worth adding. Yet if an admissible intervention targets the composite directly, say by perturbing one subsystem while leaving the driver intact, the previously synchronized trajectories decouple and the composite's macro-transition structure breaks down. The predictive sufficiency was parasitic on the common cause, not grounded in autonomous transition organization within the composite itself. Observational diagnostics alone cannot distinguish this case from genuine closure; the intervention-indexed criterion can.

The present criterion delivers a different verdict precisely because it indexes objecthood to intervention-stability, not to observational compression alone.

This is also the point of nearest disagreement with Franklin and Robertson. Their screening-off plus novelty condition is an important admittance filter, and the case just described is not meant to trivialize it. The point is narrower. Their criterion asks whether the macro-variable enters into a genuine macrodependency that renders some lower-level detail conditionally irrelevant while contributing something novel at the macro level, what they call a "screening off condition as well as novelty" (Franklin and Robertson 2024, p. 1). The common-cause proxy is introduced as a pressure case for exactly that package. Under passive observation, the proxy inherits a stable macrodependency from the upstream driver. Relative to a target explanandum, conditioning on the proxy can then render a wide range of lower-level detail irrelevant in just the way Franklin and Robertson emphasize, because the relevant dependence is borrowed into the proxy by the shared cause. The case therefore deserves to be treated as a genuine pressure case for the screening-off package rather than as a verbal trick. It presents stable macrodependence and apparent novelty at the target grain while still leaving open whether the proxy itself carries autonomous transition structure.

Closure addresses that remaining question. If an admissible intervention perturbs the proxy while leaving the common cause intact, the apparent autonomy disappears.

The macro-variable was not organizing its own transitions; it was borrowing its stability from upstream structure. What the case shows, then, is not that screening off is spurious or dispensable, but that screening off by itself does not yet settle the issue of autonomous macro-objecthood.

That further point matters for the paper's larger metaphysical claim. Once closure is added, the rainforest is not merely the permissive thought that many levels may be real. It becomes a disciplined space of admissible coarse-grainings. Rosas et al. supply formal tools for identifying relations among such partitions (Rosas et al. 2024). What the present paper adds is the ontological reading: the rainforest's structure is the lattice of computationally closed coarse-grainings, within which admission, qualification, and downgrade can be reported explicitly.

This is why the present paper treats the rainforest-admittance protocol as staged. Compression identifies candidates. Screening off and novelty remove obvious gerrymanders. Closure then determines which survivors carry genuinely autonomous transition structure. The stages are cumulative rather than rival. The disagreement is only that screening off cannot do the final sorting by itself.

A similar point applies to causal-emergence work. Effective-information analyses identify when macro-descriptions gain determinism relative to micro-descriptions (Hoel et al. 2013); Dewhurst (2021) argues that such gains support only epistemic emergence absent further constraints. That diagnosis is correct as far as it goes, but it identifies a gap rather than filling it. A partition can increase effective information while remaining fragile under intervention or horizon variation. The present approach treats such gains as corroborating evidence within a closure-governed criterion, not as a standalone ontology test.

Relatedly, Wallace's dynamic treatment of higher-level ontology is right to push the debate away from mere collections of parts and toward structure "in the distributions and dynamics" (Wallace 2024). But a commuting or phase-space friendly representation is still not yet a full admittance criterion. Closure asks when the higher-level structure is stable under admissible perturbation rather than merely calibrated to observed trajectories. Ladyman's recent contribution to the real-patterns volume moves in the same dynamic direction. He characterizes real patterns in terms of "projectible generalisations that predict and explain" (Ladyman 2026, p. 1). The present proposal is that what remains to be added is explicit admission and downgrade discipline.

So the paper is not computational mechanics or causal-emergence analysis with new labels. It is a philosophical criterion that borrows formal tools while adding intervention-indexed objecthood conditions, an explicit commitment protocol, and a sustained argument about why closure is the right fix for pattern-realism's permissiveness problem.

4 Approximate Closure Without Instrumentalist Drift

4.1 Why Approximation is Expected

In complex systems, boundaries leak and couplings shift across regimes. Exact closure is therefore unusual. A credible realism criterion cannot require perfect closure everywhere. The study of approximate aggregation has a long history, especially Simon's account of hierarchy and near-decomposability, where strong within-subsystem couplings can coexist with weaker cross-subsystem couplings and thereby separate short-run local dynamics from longer-run aggregate behavior (Simon 1962; Ando and Simon 1961). The present extension is philosophical rather than technical: it uses graded closure to support graded ontological verdicts, not to improve estimation. Rosas et al.'s distinction between informational, causal, and computational closure is useful here because it shows why the formal story is already graded and plural at the diagnostic level even when the present paper uses it to motivate one ontological criterion (Rosas et al. 2024).

Approximation here is not concession to arbitrariness. It is the expected non-ideal form of the same criterion, provided constraints are fixed and diagnostics are comparative.

The right ontology test is type-level before token-level. A pattern type can be robustly closed for a regime even when a specific token fails because of boundary violation, atypical perturbation, or timescale mismatch. One anomalous token is therefore not decisive. The relevant question is whether failures are exceptional or systematic for the type under the stated constraints. Systematic failure, where within-class micro-differences repeatedly matter for macro-transitions across tokens under fixed constraints, is what triggers type-level downgrade rather than token-level exception.

4.2 Leakiness as Canonical Target

Leakiness measures how much within-class micro-detail still improves macro-future prediction once current macrostate is fixed. A canonical quantity is the conditional mutual information $I(X_t; Z_{t+1} \mid Z_t)$ (Cover and Thomas 2006, eqs. (2.60)–(2.61)). Low values support closure. High values indicate hidden transition-relevant heterogeneity.

Using information-theoretic measures to define ontology might appear to assume a computationalist metaphysics, but the justification is strictly physical. Information processing incurs thermodynamic costs (Landauer 1961). A system, whether a biological organism or an engineered control mechanism, that tracks full micro-details without gaining macro-predictive advantage incurs unsustainable energetic overhead. Evolution and physical self-organization therefore robustly select against leaky boundaries.

Measuring leakiness is not an imposition of computer-science metaphors onto nature; it is a formal accounting of the physical and thermodynamic constraints that force systems to dimensionally reduce their own state spaces. A low-leakage partition tracks the precise joints where the physical world successfully minimizes this metabolic and computational cost.

In this paper, leakiness is the default target quantity. Other diagnostics, such as within-class transition divergence and predictive gain from added micro-features, function as estimators or proxies when direct estimation is limited.

Leakiness should therefore be read comparatively and through robustness checks, not as a context-free absolute cutoff. Absolute tolerance is domain-relative, but high sensitivity of leakiness rankings to modest defensible modeling choices is itself a downgrade trigger.

Comparative ranking is necessary but not sufficient. A candidate can beat nearby alternatives and still fail objecthood if absolute leakiness remains high and instability persists under modest robustness checks. The criterion therefore requires both relative superiority and minimum viability.

The viability floor can be stated without a numeric threshold. A partition fails minimum viability when, across the declared robustness checks, within-class micro-features yield consistent, non-negligible gains in macro-future prediction or intervention response relative to the best macro-only model. Persistence of that pattern across diagnostics and perturbation tests is decisive: the partition has not achieved the transition autonomy the criterion requires, regardless of how it ranks against competitors.

The term "non-negligible" can be given a comparative anchor without fixing a universal numeric threshold. Leakiness is most clearly non-negligible when adding within-class micro-features yields prediction gains comparable in magnitude to the gains the macro-partition itself provides over a naive baseline model. If the macro-partition delivers substantial predictive improvement over no partition at all, and within-class micro-features add only marginal further gain, that residual leakiness is negligible for most objecthood purposes. If micro-features instead add gains approaching or matching the macro-partition's own contribution, the partition is not doing most of the relevant work, and minimum viability is not met. This framing makes the judgment about a single tractable quantity, namely relative predictive contribution, rather than an abstract absolute cutoff, and it makes cross-domain comparisons more tractable. The two requirements work in sequence: the viability floor is applied first as a minimum gate, excluding candidates regardless of how they compare to rivals, while comparative ranking then assigns verdict categories among candidates that pass it. To keep this anchor non-discretionary, the baseline model class and gain metric must be fixed upstream with regime and admissibility, and post hoc revisions trigger restart and disclosure.

Verdict-category boundaries carry genuine vagueness because the underlying property of transition autonomy is itself graded. This is a feature rather than a defect: biological fitness is real and explanatorily significant without admitting a sharp boundary between fit and unfit. What matters primarily is ordering stability. When partition A consistently shows lower leakiness and greater perturbation stability than partition B across diagnostics, that ordering should be robust even if the exact placement of the boundary between "qualified" and "robust" carries some domain-conventional element. A partition that dominates alternatives across diagnostics and remains stable under modest changes to horizon, intervention distribution, and robustness checks earns the stronger verdict regardless of where one draws the categorical line.

4.3 Procedural Safeguards in Non-Ideal Cases

Approximate closure claims require explicit safeguards.

1. Fix regime, horizon, admissibility class, and diagnostics before comparative scoring.
2. Compare candidate partitions under the same fixed setup.
3. Test robustness under modest changes in horizon and intervention distribution.
4. Apply a minimum-viability floor: if all candidates remain high-leak and fragile, return no robust objecthood verdict for that regime.
5. Report verdicts as robust, qualified, or indeterminate.

These safeguards also fit Wimsatt's robustness-analysis idea: confidence should rise when partially independent ways of detecting the same structure converge, and it should fall when apparent agreement is driven by shared assumptions or disappears under small perturbation (Wimsatt 1981). Multiple diagnostics do not replace criterion-level argument with a vote count. They reduce the risk that a favorable verdict is an artifact of one estimator, one horizon choice, or one admissibility specification.

This keeps threshold choice procedural rather than discretionary. It also addresses a predictable confusion. Estimation difficulty is an epistemic limit on access, not a defect in the metaphysical criterion itself.

Selection discipline. One concern from both reviewers and readers is that flexibility can silently re-enter through upstream choices. The answer is a single discipline rule: admissibility, state construction, horizon, and diagnostics are fixed before scoring, and post hoc revisions trigger restart and disclosure. This does not eliminate judgment. It makes judgment auditable and prevents favored partitions from being protected by moving criteria.

4.4 Regime Dependence and Projectibility

Regime dependence does not imply observer-relativity. It states that closure depends on actual transition structure under specified constraints. A pattern can be closed in one regime and leaky in another because the structure has changed, or because the available control channels have shifted. Since the admissible intervention class is defined by the regime's transition structure (§2.5), a change in that structure can alter both the admissible class and the closure verdict even when the system's microdynamics remain the same.

The metaphysical stance is explicit: regime and horizon index the modal profile being claimed, not a concession that anything goes.

Projectibility provides a further check. Defensible regime specifications should support stable induction under modest counterfactual variation. Narrowly engineered regimes that protect a favored partition usually fail under small context shifts. When that occurs, the candidate was never robustly closed in the relevant sense.

This is where the paper is closest to Ladyman and Ross. Projectibility is not treated as an optional methodological virtue. It functions as a realism-relevant stress test on whether a closure claim survives beyond calibration conditions (Ladyman and

Ross 2007). Closure without projectible robustness is too cheap to support robust objecthood claims.

Ladyman and Lorenzetti's effective ontic structural realism sharpens the same point at the metaphysical level. If ontology is regime-dependent and scale-relative, then the question is not whether a candidate belongs to a single final inventory of being, but whether it is stable enough in the relevant regime to merit commitment there (Ladyman and Lorenzetti 2024). Closure is offered here as the operational discipline for making that judgment explicit.

Regime dependence itself should be split into two forms. Ontic regime dependence occurs when system structure changes, such as phase transitions or boundary-condition shifts. Epistemic regime dependence occurs when measurement or control access changes while underlying structure remains fixed. The first changes what is there to be tracked. The second changes what can be warranted from available evidence.

4.5 One Distributed Illustration

Macro does not mean large or spatially contiguous. Coarse-graining can be logical and distributed. Monetary systems illustrate this point without requiring new machinery.

The macro-transition structure of transactions can remain stable across heterogeneous micro-realizations, such as cash tokens, ledger records, and digital balances, under admissible legal and financial interventions. If that stability holds, macro-objecthood is warranted in regime. If implementation channels degrade, closure can fail even while symbolic representations persist.

A minimal failure case clarifies the point. A state can retain formal legal tender rules while losing reliable payment enforcement and settlement implementation. Closure fails not because the rules changed but because the physical infrastructure that enforced screening-off is gone. The representation persists but transition autonomy degrades, and closure-based commitment should be downgraded. The same structure applies more broadly: the criterion explains when and why a social entity stops being a real macro-object without requiring the claim that it was never real, or that it remains real simply because people believe in it. On the broader rainforest realist picture, the relevant issue is whether transactional organization still supports the economic structure in question, not whether the label survives institutionally (Ladyman and Ross 2007).

Organisms give a complementary illustration. In many physiological regimes, membrane and regulatory organization screen off large amounts of molecular variation for the target transitions, so intervention on organ-level variables remains predictively and manipulatively effective. When those screening structures fail, for instance under severe systemic breakdown, the same macro-description can become leaky and require finer-grained tracking. The point is not that organism talk is always closed. The point is that closure can be regime-robust without being regime-universal.

Across these illustrations, the verdict structure does real discriminative work. The monetary example in a well-functioning system approaches robust objecthood in the declared regime: macro-transition structure remains stable across heterogeneous realizations and admissible interventions. The organism example falls into the qualified range: within established physiological regimes, membrane organization and regulatory control screen off large amounts of molecular variation, making organ-level interventions reliably effective, but leakage appears at ecological or pathological boundaries where those screening structures degrade. The failed-implementation case represents a downgrade: formal representations persist while transition autonomy erodes, warranting withdrawal of robust commitment. The three cases illustrate the verdict categories in operation, not as abstract labels but as outcomes determined by the same diagnostic criteria applied to different evidential situations.

The illustration does one job only. It shows that closure tracks transition structure, not spatial shape. Nothing in the argument depends on taking a stance on broader social ontology.

4.6 Failure Conditions and Downgrade Rules

The account should also say clearly when commitment should be withdrawn or downgraded. Four failure patterns are especially relevant.

1. Persistent high leakiness across defensible diagnostics under fixed constraints.
2. Strong disagreement among diagnostics that does not resolve under modest robustness checks.
3. High sensitivity of verdicts to small, defensible changes in admissibility or horizon.
4. Candidate sets where no partition clears the minimum-viability floor.

When these patterns appear, the right response is not forced binary judgment. The right response is explicit downgrade to qualified or indeterminate status. This keeps the criterion resilient without pretending that every case must produce a sharp ontology verdict.

Conversely, when closure persists across distinct, independently motivated admissibility classes within the same regime, that cross-admissibility stability raises evidential confidence for robust commitment.

Type-token reminder: a downgraded token does not automatically refute type-level closure. The key question is whether instability is exceptional at token level or systematic for the type under the stated constraints.

4.7 Stable Versus Merely Entailed Patterns

The graded account also supports an important distinction. A pattern can be entailed by microhistory and laws without being stable as a macro-handle. Entailment alone is cheap. Stability under admissible perturbation is demanding.

This distinction matters for permissiveness debates. A contrived disjunctive construction can be true of a realized trajectory while still failing closure. It can require continual within-class micro repair, fail under modest regime shifts, and lose interventional reliability.

A historical case makes the pattern concrete. Phlogiston theory compressed combustion phenomena under a single macro-variable, but the partition required persistent bookkeeping to survive: when metals gained weight upon burning, theorists introduced “negative phlogiston”; when different substances showed different weight changes, further ad hoc parameters appeared. Each patch was a new within-class distinction imported to preserve macro-prediction. By contrast, oxygen theory required substantially less such repair: the macro-variable (oxidation state) screened off the relevant chemistry without persistent bookkeeping. The critique here is structural, not retrospective ridicule. A candidate that continually imports corrections to preserve macro-prediction behaves like a non-autonomous partition, and the closure framework detects this failure directly.

So the claim is not that gerrymandered constructions are false. The claim is that they usually fail to qualify as macro-objects. They may remain descriptions, but not object-level descriptions in the relevant regime. The same applies to computationally adequate models more broadly. A higher-level model can be useful for prediction or control without meeting closure standards. Computational adequacy means the model serves some purpose; closure requires that transition-relevant micro-differences are screened off in the specified regime. A computationally adequate but high-leak representation remains a valuable tool, but it does not automatically earn macro-object status.

4.8 Practical Qualification Under Sparse Intervention Access

In many domains, direct interventions are sparse, ethically constrained, or expensive. This does not invalidate the criterion. It changes evidential strategy. The evidential order is asymmetric: predictive success can raise confidence, but only interventional stability can underwrite objecthood commitment.

Where intervention data are limited, predictive closure functions as an operational indicator while interventional closure remains the ontological target. A model can fit historical trajectories and still fail under novel perturbation. For that reason, claims should be qualified by available intervention access and downgraded when out-of-regime behavior is unstable.

In-sample prediction is therefore not treated as decisive. The standard remains robustness under admissible perturbation, even when that robustness must be assessed indirectly.

This point also explains why diagnostics should be triangulated. Predictive closure can look strong in-sample while interventional closure fails under admissible perturbation. Partial observability can hide within-class heterogeneity that later appears as instability. Nonstationarity can make a previously low-leak partition unstable outside its calibration window. Triangulation does not remove these risks, but it makes them visible.

4.9 Conceptual Contrast: Near-Tie Prediction, Different Ontology Verdicts

Two partitions can show similar short-horizon observational performance and still receive different closure verdicts. This is where the criterion does real philosophical work.

Suppose partition P_1 and partition P_2 produce comparable one-step observational fit in calibration data. Under fixed admissible interventions, P_1 preserves stable macro-transition structure while P_2 requires repeated within-class micro refinements to maintain performance. Observationally, they may look close. Structurally, they are not.

On the present account, P_1 receives the stronger objecthood verdict because it remains transition-autonomous under the fixed constraints. P_2 can remain useful as a representation, but its dependence on recurring hidden repair counts against macro-object standing.

A concrete sketch helps. In traffic modeling, one partition may track density and flow by lane segment. A rival partition may track arbitrary vehicle-ID clusters that happen to fit one week of observations. In-sample one-step fit can be similar. Under admissible perturbations, such as ramp metering changes and speed-limit adjustments, the lane-flow partition remains stable while the ID-cluster partition becomes brittle. Under modest horizon extension, leakiness for the ID-cluster partition rises sharply. The verdict is then robust or qualified objecthood for the lane-flow partition and downgrade for the rival.

Mini-walkthrough under the reporting schema:

1. Target partitions: P_{lane} (lane-segment density and flow) versus P_{id} (vehicle-ID clusters).
2. Regime and horizon: weekday commuter traffic in one metropolitan corridor; horizon set to short-run control windows and one modest extension beyond calibration.
3. Admissibility class: control channels that traffic operators actually use in that regime (ramp metering and speed-limit controls), excluding interventions that require implausible vehicle-level actuation.
4. Diagnostics: predictive gain from within-class microfeatures, within-class transition-profile divergence, and intervention response invariance under admissible perturbations.
5. Verdict: P_{lane} remains comparatively stable and earns robust or at least qualified objecthood in the declared regime; P_{id} remains instrumentally useful in calibration but is downgraded once perturbation and modest horizon extension are applied.

A second illustration shows the criterion sorting between two scientifically rigorous partitions. Consider two ways to divide an organism into macroscopic parts. One partition, P_{organ}, follows functional physiological boundaries: heart, lungs, liver. A rival partition, P_{slice}, divides the body into strict transverse geometric planes, as used in medical tomography. Under resting observation, both achieve comparable short-horizon compression of local mass, fluid density, and electrical activity. Under an admissible physical perturbation, such as cardiovascular drug administration or sudden exertion, their transition autonomy diverges. P_{organ} preserves its macro-transition structure: the heart responds as a coordinated unit and its macroscopic variables remain sufficient for predicting onward causal trajectory. P_{slice} fractures, because functional tissues span across geometric planes; predicting the macro-future of a given slice now requires tracking the specific intercellular signals and fluid volumes crossing its boundary. On the present view, P_{organ} earns robust objecthood in the physiological regime because it isolates transition-autonomous parts; P_{slice} remains a valuable diagnostic tool but its recurring dependence on hidden micro-state repair prevents it from qualifying as a macro-object under the declared constraints.

4.10 Reporting and Evidential Discipline

To keep conclusions comparable across cases, closure assessments should report five items explicitly.

1. Target partition and rationale.
2. Regime and horizon specification.
3. Admissible intervention class and justification.
4. Diagnostics used and their agreement or disagreement.
5. Final verdict category: robust, qualified, or indeterminate.

This reporting structure is simple, but philosophically important. It prevents silent shifts in target, timescale, or admissibility from being mistaken for genuine ontological progress.

It also keeps the verdict vocabulary tied to ontology rather than to mere confidence. A robust verdict reports that a candidate occupies a stable place in the relevant closure structure under modest checks. A qualified verdict reports a real but fragile place in that structure. An indeterminate verdict reports that no stable placement has yet been secured. The reporting schema is therefore how a lattice-shaped ontology is made explicit in practice. Rosas et al. provide the formal architecture for talking about nested closure structure; the present paper's contribution is to turn that architecture into an ontological reporting discipline (Rosas et al. 2024).

For consistent application, the assessment sequence should be explicit.

1. Specify candidate partitions for one target process.
2. Fix regime, horizon L, admissible intervention class $\mathcal{I}$, diagnostics, and model classes in advance.
3. Evaluate all candidates comparatively under that same fixed setup.
4. Report verdict category and sensitivity under modest perturbation checks.

This sequence is not an optional implementation preference. It is part of what makes closure claims epistemically disciplined rather than post hoc.

Diagnostics do not replace criterion-level argument. They provide evidence about whether a candidate satisfies the criterion under declared constraints. This distinction matters because reviewers can otherwise misread the proposal as reducing ontology to whichever metric happens to be available. The evidential logic is comparative and convergent. No single proxy is treated as infallible. Confidence increases when different diagnostics track the same rank-order among candidate partitions and when those rankings remain stable under modest perturbation tests. Confidence decreases when diagnostics diverge persistently or when rankings are brittle under small design changes.

Verdict categories are graded accordingly. Robust verdicts require convergence and stability. Qualified verdicts fit mixed but non-trivial evidence. Indeterminate verdicts fit persistent instability. The diagnostic set can vary by domain, but it should answer one fixed question: do within-class micro-differences still matter for macro-what-follows? In practice, three checks are usually enough.

1. Leakiness proxy based on predictive gain from added within-class micro-features.
2. Within-class transition-profile divergence.
3. Intervention-response invariance under admissible perturbation.

Agreement across these checks increases warrant. Persistent disagreement is evidence for revision or downgrade, not for forced commitment. When interventional evidence is sparse or unavailable, the remaining diagnostics can still support qualified status, but robust objecthood requires that the interventional leg not be entirely missing.

Minimal triangulation recipe: require stability under modest horizon variation and under at least two defensible admissibility perturbations, while at least two diagnostics agree on candidate rank-order. If this condition fails, return qualified or indeterminate status rather than robust objecthood.

5 Objections and Replies

The objections are organized as a progression from foundational worries to evidential-discipline worries. The order is deliberate, but the core answer remains stable: closure under fixed constraints supplies ontological discipline without demanding a single privileged descriptive level.

Three standing commitments apply throughout the replies below and will not be restated each time. First, the account does not deny microphysical completeness; it claims non-redundancy at the explanatory and interventional level for declared macro-targets. Second, all verdicts are indexed to the declared constraints, so objections that presuppose context-free ontological claims address a view the paper does not hold. Third, the strongest conclusion is conditional on structural realist and interventionist commitments; readers who withhold those commitments can still accept the narrower anti-gerrymandering result.

5.1 "Beyond Screening Off"

Objection: Franklin and Robertson already provide a principled admittance criterion for rainforest realism. If emergence requires screening off plus novelty, one might wonder what substantive work is left for closure to do (Franklin and Robertson 2024).

Reply: the paper accepts screening off and novelty as indispensable, but denies that they are sufficient. Their criterion asks whether the macro-variable makes micro-details conditionally irrelevant for the relevant explanandum while contributing a genuine macro-level dependency. Closure asks a further question: does the candidate macro-variable carry its own transition structure under admissible intervention, or was the screening off borrowed from elsewhere?

The common-cause case from §3.3 is the wedge. Under passive observation, a proxy variable can inherit a stable macrodependency from an upstream driver and can thereby make a range of micro-detail conditionally irrelevant relative to the target explanandum. That is why the case is a genuine pressure case for the screening-off package rather than a merely verbal counterexample. The screening-off appearance is not fictitious. It is borrowed. If an admissible intervention perturbs the proxy while leaving the driver intact, the apparent autonomy disappears. The macro-variable did not carry autonomous transition organization after all. Closure therefore sorts one class of survivors that screening off alone does not yet exclude.

This is not a hostile revision of Franklin and Robertson's program. The present paper relies on it. Their criterion removes obvious gerrymanders and clarifies the importance of law-like macrodependence. The present claim is simply that one further filter is needed for robust admission to the rainforest. Compression identifies candidates. Screening off plus novelty narrows the field. Closure then determines which survivors remain autonomous under admissible perturbation. That is best understood as a completion of the admittance protocol, not a replacement.

5.2 "Instrumentalism and Admissibility"

Objection: target selection is interest-relative, so the view still collapses into usefulness.

Reply: target selection can be interest-shaped without making closure verdicts preference-shaped. Once regime, horizon, and admissibility are fixed, whether macro-transitions are autonomous is a system fact under explicit constraints.

The stronger version invokes Dennett's Martian (Dennett 1991, p. 42). If an ideal micro-predictor can forecast everything, macro realism looks optional. The mistake is to infer ontological redundancy from representational power. Even a micro-omniscient predictor coexists with objectively better macro-bottlenecks for target dynamics. Ignoring those bottlenecks increases representational burden without erasing macro transition structure.

The deeper point is that observer power changes convenience, not closure facts. Whether a partition is transition-autonomous under fixed constraints does not vary with who computes it.

Admissibility version of the same objection: admissibility criteria already encode what counts as the relevant level. The reply appeals to hierarchical order and disclosure rules. Admissibility is fixed by physically realizable, regime-preserving control channels before partition scoring. Post hoc admissibility revision triggers restart. This makes back-fitting explicit and sanctionable.

Residual disagreement can remain. The account does not promise universal convergence from one pass. It promises disciplined comparison and explicit downgrade when verdicts are unstable across nearby defensible admissibility specifications. This is a feature for skeptical readers: hard cases are flagged as unresolved rather than settled by stipulation.

Instrumentalist pressure hides in two forms: ontology should track predictive utility only, or ontology is unnecessary once predictive utility is available. The present view rejects both. Utility without closure is too permissive, and closure without ontology understates what stable intervention-guiding structure provides. What closure adds is not metaphysical extravagance. It is commitment discipline. If a candidate repeatedly supports stable counterfactual control under fixed constraints, refusing any ontological standing starts to look like an empty verbal policy rather than a substantive alternative. The paper does not force maximal realism. It forces explicit criteria for when anti-realism remains credible.

The strongest anti-instrumentalist point is structural rather than rhetorical. The objection that leakiness is "only" predictive error presupposes a gap between tracking causal structure and describing what is real. But under the paper's structural realist commitments, there is no further ontological layer for that gap to open onto. Stable counterfactual and predictive organization is not evidence for some deeper objecthood condition left unspecified elsewhere; it is what the relevant reality consists in at that grain. A high-leak partition is not one where we failed to see the real object hiding underneath. It is one where the candidate genuinely lacks autonomous transition structure. Leakiness therefore measures ontic deficiency in the candidate partition, not merely our ignorance. Once the test and the target are properly separated, the prediction miss is itself the structural fact.

This is also where the cheap-coding objection is handled directly. A recoding can improve local fit and still fail closure under admissible perturbation. The criterion is therefore representation-hostile in exactly the way permissiveness critiques demand.

5.3 "Causal Exclusion Still Defeats Macro Claims"

Objection: if microphysics is causally complete, macro-level causal claims are redundant (Kim 1998).

Reply: closure does not posit a second fundamental cause layer. It identifies when macro-description is sufficient for prediction and intervention at the relevant target level. Microphysical implementation remains untouched.

The standard reply to Kim's exclusion argument is that macro-causation is explanatorily indispensable. The closure criterion allows a stronger, structural reply: macro-states are non-redundant because they filter out causal degeneracy. Due to multiple realizability, myriad distinct micro-configurations map to identical macro-outcomes. Tracking every electron is not merely computationally expensive; it actively introduces non-difference-making noise into the causal model. When a system achieves closure, the macro-boundary strips out this micro-level degeneracy, effectively concentrating causal power. The macro-variable is therefore not a convenient shorthand competing with a fundamental micro-cause; it is a more informative causal parameter for the target outcome because it isolates the relevant difference-maker. Supervenience establishes a dependency relation, not an identity relation, and causal priority follows the level where conditional independence actually screens off this noise.

This reply also handles the fat-handedness worry that macro-interventions inevitably co-intervene on micro-states (Dewhurst 2021). Co-intervention on the micro-level is real, but closure ensures that the within-class micro-differences it produces do not change macro-transitions; the co-intervention is therefore non-difference-making for the macro target. Macro-level interventions can be successful and projectible across perturbations without denying microphysical realization (Woodward 2003; Pearl 2009; Hoel et al. 2013). Compatibility with microphysical completeness is a design feature, not a concession.

The overdetermination worry can be handled directly. The proposal does not claim two independent sufficient causes for one event. It claims one implemented process with multiple adequate descriptions at different explanatory and control targets. Exclusion pressure weakens once adequacy is indexed to target and intervention class rather than to a single privileged level.

Relative to Kim's strongest formulations, the key move is target-indexing rather than layer multiplication (Kim 1998, 2005). A closed macro-description can be non-redundant without competing for micro-level fundamentality, because non-redundancy is inferred when micro-detail ceases to add control-relevant information for the target outcome under fixed constraints. This is not a linguistic escape. It is a claim about which counterfactual dependencies remain stable for a declared class of interventions. Compatibility with microphysical completeness by itself is too weak to do this work, because it leaves open whether macro-variables are dispensable shorthand. Closure narrows that space: when it holds, dispensability must be argued against a stated record of interventional and predictive sufficiency, not simply asserted.

5.4 "Autonomy, Pluralism, and Distinctiveness"

Objection: the view sounds like a formal restatement of familiar special-sciences autonomy claims, not a distinctive realism thesis.

Reply: the paper is continuous with autonomy insights, but it is not equivalent to them. Generic autonomy claims often remain permissive about what qualifies as a level-worthy grouping. The present view adds a discriminating closure condition with explicit exclusion and downgrade rules. That addition changes verdict structure.

The difference is practical and philosophical. Practical, because the account gives a protocol for ruling out high-maintenance, transition-incoherent candidates. Philosophical, because it ties realism commitment to transition autonomy under admissible interventions rather than to explanatory convenience alone. In this respect, causal-emergence style results are treated as evidence within a closure-governed approach, not as a replacement for the criterion itself (Hoel et al. 2013).

The same comparison can be made with the mechanisms literature. Mechanistic accounts are right to tie level talk to organized structure that supports stable intervention and explanation (Machamer et al. 2000; Craver 2007). Recent structuralist work on mechanisms makes the same pressure explicit: object-based metaphysics for mechanisms faces the problem of fitting special-science ontology to physics without collapsing it into mere convenience (Jiang 2024). But even that move does not by itself settle which variables or entities have been carved at the right grain for the target explanation. Closure addresses that earlier question by testing whether a candidate macro-variable screens off the micro-differences that would otherwise keep the mechanism explanatorily open.

The view should therefore be read as a constrained realism refinement of autonomy views, not as an unrelated alternative and not as a mere restatement. The conceptual difference can be stated directly. Generic autonomy claims assert that higher levels matter. Closure specifies *when* they matter, *how much* they matter (graded by leakiness), and *when the claim should be withdrawn* (explicit downgrade conditions).

A realism criterion without withdrawal conditions is not a criterion at all. It is an assertion.

A more specific comparison is worth drawing. The classical arguments for the autonomy of the special sciences established that higher-level kinds can be explanatorily indispensable precisely because they abstract over heterogeneous micro-realizations (Putnam 1967; Fodor 1974). Multiple realizability is in fact a structural feature of any genuinely closed macro-kind: if a partition closes, different microstates within the same macroclass by construction produce the same macro-transitions, making the kind multiply realized in the relevant sense. Multiple realizability is therefore necessary for closure. But it is not sufficient. A purely extensional claim about kind membership says nothing about the dynamical coherence of the realizers. A disjunctive aggregate can have heterogeneous realizers while completely failing closure, because those realizers can have utterly divergent transition profiles. What closure adds is the dynamical requirement: the realizers must not only fall under the same label but must produce the same macro-transitions under the declared constraints. This gap, between what a kind groups together extensionally and how those instances behave dynamically, is precisely where gerrymandered kinds exploit multiple-realizability intuitions without earning genuine macro-object status. The present criterion formalizes the transition condition that the autonomy tradition always needed to separate genuine higher-level kinds from merely disjunctive aggregates (Shapiro 2000). The monetary illustration in §4.5 provides a concrete instance: coins, ledger entries, and digital balances are heterogeneous realizers, but what earns money macro-object status is not the heterogeneity of realizers but the transition-equivalence of those realizers under admissible financial interventions.

Pluralism version of the objection: if multiple grains can satisfy closure, ontology becomes permissive again. The reply is that plurality of closed grains does not imply arbitrariness. Once regime, horizon, and admissibility are fixed, which partitions close is determined by transition structure, not by analyst choice. When multiple candidates remain viable, robustness and minimality rank them as an objective relation among candidates under fixed constraints. In many domains these relations are further structured by coarse-graining order: when one closed partition is a refinement of another, their relation is nesting, not rivalry, which disciplines pluralism rather than dissolving it.

Andersen's density-of-structure point helps explain why this is not an embarrassment (Andersen 2025). If, as he puts it, "the world may be dense with structure," then one should expect more than one admissible closed grain in some systems (Andersen 2025, p. 1). The philosophical task is not to force uniqueness where the dynamics do not supply it. It is to report where in the closure lattice each candidate stands, how stable that standing is, and which grains are more robustly supported under the declared constraints.

This is a virtue for complex systems, not an embarrassment. Forcing a single grain in every context would be a stronger and less defensible claim than the paper needs.

5.5 "The Markov Template is Too Restrictive"

Objection: strong lumpability presupposes first-order Markov microdynamics and excludes memory-dependent systems.

Reply: the criterion is transition sufficiency, not first-order Markovity. Strong lumpability is an exact benchmark. In memory-bearing systems, state can be enriched with relevant history and the same closure question can be asked over that enriched state. This changes state representation, not criterion content.

Concrete cases make this less abstract. Immune response modeling often depends on historical exposure profiles, and institutional dynamics often depend on path-dependent enforcement and trust trajectories. In both cases, a memoryless state is too thin, but a minimally history-enriched state can still support closure testing at the target grain.

Minimality still matters. Enrichment should be the least extension needed to preserve closure performance under fixed constraints. The account also accepts an important edge case: if the minimally sufficient enriched state approaches micro-level complexity, then closure may be recovered without meaningful compression. In that regime, the result is not robust macro-objecthood. It is a warning that no informative macro-partition has been found at the target grain.

5.6 "Formal Closure Collapses the View into Formalism"

Objection: formal systems are closed by stipulation, so closure cannot ground empirical objecthood.

Reply: stipulated and induced closure must be separated. Formal systems can have objective internal closure under constitutive rules. The present criterion concerns induced closure in implemented spatiotemporal dynamics. Internal formal coherence does not by itself settle empirical macro-objecthood claims.

This does not reduce formal systems to arbitrary invention. Formal systems are invented, but not freely. What counts as intelligible formal practice is constrained by stable identity conditions, compositional inference, and coherent consequence. Those constraints explain why formal work can guide empirical reasoning without itself deciding empirical objecthood.

No collapse follows. The paper's argument is narrower and clearer: induced closure is the objecthood criterion for implemented macro-patterns in regime.

5.7 "Closure Is Too Strong and Eliminates Most Special Sciences"

Objection: if closure requires that within-class micro-differences not matter for macro-transitions, then most special-science kinds will fail. Biological species, psychological states, and economic categories are notoriously leaky. The criterion eliminates the ontology it was supposed to discipline.

Reply: this objection conflates robust objecthood with any objecthood verdict at all. The account provides three verdict categories, not one. Robust objecthood requires stable closure under fixed constraints. Qualified objecthood fits cases where closure is partial, regime-limited, or supported by convergent but incomplete evidence. Indeterminate status fits cases where evidence is too unstable to warrant commitment.

Most special-science kinds fall into the qualified range, and that is the correct result. Biological species exhibit substantial transition autonomy within ecological and evolutionary regimes while leaking at boundary cases (hybrid zones, ring species, horizontal gene transfer). Psychological kinds often support predictive and interventional regularity within clinical or experimental regimes while failing under broader perturbation. The criterion does not eliminate these kinds. It gives them the status their closure evidence supports: qualified objecthood with explicit regime limitations, rather than either full robust status or wholesale elimination.

This is also where nearby real-pattern work in biology is useful as a foil. One can characterize biological objects by their explanatory power and pragmatic utility, and by the availability of multiple valid carvings, without yet having shown that lower-level variation is transition-irrelevant in the stronger sense used here (Jiang 2025). The present paper does not deny that such utility matters. It denies that utility by itself settles objecthood.

The objection therefore proves too much. A criterion that granted robust objecthood to every special-science kind regardless of leakiness would be the permissive view this paper is designed to replace. The gain from graded verdicts is that they match evidential reality rather than forcing a binary choice between full realism and eliminativism.

5.8 "Predictive Success and Failure Conditions"

Objection: a model can predict well without warranting ontological commitment.

Reply: that is correct, and the account does not deny it. Predictive success functions as evidence only when it aligns with closure diagnostics and interventional stability.

That is also the natural place to distinguish the present view from the strongest nearby tightening of real-pattern realism. Millhouse shows why compression, similarity, and counterfactual support already improve on a purely permissive reading of Dennett (Millhouse 2022). The present claim is simply that even those stronger constraints leave open whether the candidate partition itself carries autonomous transition structure under admissible perturbation.

This point carries two immediate consequences.

First, protocol language has a bounded philosophical role. The paper includes diagnostics to avoid a familiar failure mode, where a criterion is asserted but left too unconstrained to guide verdicts. The claim remains conceptual. Diagnostics do not replace argument. They operationalize what the argument says should matter.

Second, the criterion is falsifiable in its own terms. If candidate partitions repeatedly fail closure diagnostics across defensible regimes and admissibility classes, commitment should be withdrawn or downgraded. The same applies when verdicts are unstable under small, defensible changes in horizon, admissibility, or diagnostic proxy. A proposal that cannot risk downgrade is not a serious realism criterion.

The proposal does not promise certainty from one metric or one pass. It promises explicit criteria for when confidence increases, when confidence should pause, and when commitment should retreat.

5.9 "Horizon-Relativity Makes the Criterion Too Weak"

Objection: if closure is indexed to horizon, any candidate can be made to pass at a short enough horizon.

Reply: horizon indexing is a constraint, not an escape clause. Two requirements prevent it from becoming one.

First, horizons must be anchored in independently identified timescale structure in the target domain: known relaxation windows, intervention latency, control-response periods, or similar physically motivated temporal features. A horizon chosen only because it makes a favored partition look closed has no independent standing.

Second, verdicts must survive a modest-extension stability check. A candidate that passes at horizon L but fails immediately at $L + \Delta$ for small, defensible Δ receives, at best, qualified status. Robust objecthood requires that closure not be razor-thin in temporal scope.

Together, these requirements force explicit timescale discipline and prevent silent switching between incompatible temporal targets. Horizon flexibility is acceptable when it tracks independently motivated structure. It is not acceptable when it is introduced only after score inspection to salvage a preferred partition.

This distinction is central for pre-emption. Critics are right that any criterion can be trivialized if timescales are unconstrained. The present account avoids that outcome by tying horizons to ex ante justification and by requiring modest-extension robustness. If robustness fails, the verdict downgrades. The criterion does not break. It reports that the claim was too strong for the evidence.

5.10 "The Criterion Is Too Conservative"

Objection: by demanding closure and downgrade discipline, the criterion may be too conservative and may miss emerging macro-objects.

Reply: conservatism is partly intentional. The paper aims to avoid premature ontological inflation. Still, the criterion is not rigid. It allows qualified verdicts for emerging patterns when closure evidence is partial but improving, and it allows verdict revision as regimes stabilize and intervention evidence accumulates.

So the criterion does not require all-or-nothing maturity before any commitment. It requires that commitment strength track available closure evidence. Woodward (2021) similarly argues that while minimal interventionist criteria establish causality, invariance is a graded virtue. Our criterion of closure can be seen as the dynamical realization of this: a macro-object is "proportional" precisely when it minimizes the omission of relevant micro-detail. This is a virtue for dynamic systems where objecthood can be developmental rather than instantaneous.

This is also where compression and closure reconnect in a non-vacuous way. Early in a domain's development, compression gains may appear before robust closure is demonstrated. The account treats this as evidential lead, not ontological conclusion.

As closure evidence accumulates, commitment can strengthen. If closure evidence fails to accumulate, commitment should remain qualified or be withdrawn even when short-run compression remains attractive. The criterion is therefore neither eliminativist nor inflationary: it does not deny standing to emerging patterns, but it does not award standing before the dynamical evidence warrants it.

6 Conclusion: A Disciplined Realism Claim

Rainforest realism is compelling, but its admittance criteria remain incomplete. This paper argues for a staged protocol. Compression identifies candidate patterns. Screening off plus novelty, in Franklin and Robertson's sense, removes obvious gerrymanders and secures serious macrodependence (Franklin and Robertson 2024). Closure then determines which survivors carry genuinely autonomous transition structure under fixed regime, horizon, and admissible intervention class. Strong lumpability provides the exact benchmark. Leakiness and convergent diagnostics extend the same criterion to non-ideal settings without changing its content.

This further step also sharpens the non-redundancy demand already implicit in the Primacy of Physics Constraint. A candidate is not non-redundant merely because it is physically compatible or descriptively useful. It is non-redundant when lower-level variation within its macroclasses stops mattering to macro-what-follows under the declared constraints. That is the sense in which closure gives rainforest admission a genuinely dynamical discipline.

What this adds to the recent literature is specific. Relative to Franklin and Robertson, it supplies a transition test for parasitic screening-off cases. Relative to Rosas et al., it turns formal closure machinery into an ontological criterion with explicit verdict and downgrade structure (Rosas et al. 2024). Relative to Ladyman and Lorenzetti's effective ontic structural realism, it offers an operational answer to when regime-dependent, scale-relative structures should be admitted, qualified, or withheld (Ladyman and Lorenzetti 2024). Relative to recent dynamic and phase-space formulations of pattern realism, including Wallace and Ladyman's contribution to the recent volume, it adds admission and downgrade discipline rather than a merely dynamic picture of higher-level organization (Wallace 2024; Ladyman 2026).

What the proposal buys is selective commitment. It supports robust commitment where closure is stable, qualified commitment in borderline cases, and explicit downgrade where verdicts depend on fragile admissibility or horizon choices. The proposal is intentionally middle-range: stronger than compression-only pattern realism because it adds exclusion and downgrade structure, weaker than a total-level metaphysics because it is restricted to induced closure in spatiotemporal systems under declared constraints.

The positive metaphysical payoff is correspondingly more explicit than a generic many-level realism. Once closure governs admission, the rainforest is best understood as the lattice of computationally closed coarse-grainings. Levels are discovered features of system dynamics, nested by coarse-graining relations, and ranked by comparative closure stability rather than by proximity to the micro-level. The verdict categories make that structure reportable: robust where placement in the lattice is stable, qualified where it is real but fragile, and indeterminate where no stable placement has yet been secured.

This is also where the view separates itself from eliminativist pressure. It does not treat higher-level descriptions as fictions by default, and it does not grant them standing by convenience alone. It asks whether they track stable transition structure under admissible intervention. If they do, withholding ontological standing requires a further argument. If they do not, realism should be downgraded rather than protected by rhetoric.

Clear limits remain. The proposal is conditional on the structural realist and interventionist commitments stated in §1; readers who withhold those commitments can still accept the narrower anti-gerrymandering result. It is not a complete metaphysics of levels and not a universal methods manual. Its contribution is a philosophical criterion with application discipline. Methodological implementation details remain downstream of this argument, and refining them does not alter the criterion's philosophical content. Future work should test the staged admittance protocol across domains by comparing candidate partitions under fixed constraints, rather than by multiplying new concepts. If rejected, it should be rejected because one rejects its stated commitments, not because circularity, instrumentalist drift, or scope ambiguity were left unresolved.

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